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Commodore 64 Emulator User Guide

STARTUP OPTIONS AND COMMAND CONSOLE REFERENCE

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Contents

Foreword

This guide describes the startup interface and the command interface implemented by the two Commodore 64 executables in EMULATORS: C64Emulator (without a local command console) and C64EmulatorC (with a local command console). It is based on the executable entry points, the C64Emulator inheritance chain, LocalConsole, and the complete C64 -> COMMODORE -> StandardCommandBuilder responsibility chain.

Command and parameter names are shown exactly as accepted by the current code. Startup options use /XVALUE with no space between the one-letter option and its value. Command names are matched as written by the builders; use uppercase names for reliable operation. Positional command parameters are separated by spaces; examples use paths without embedded spaces.

The non-console executable does not expose an interactive prompt. When started with /o it creates a remote communication system using the same C64, Commodore and standard command builders. Commands marked Local console only are intercepted by LocalConsole and therefore cannot be sent to the non-console executable.

1. Startup Parameters: Version Without a Local Console

Executable: C64Emulator.exe. It accepts the framework and C64 options below, plus /o and the communication port option. Option letters are case-sensitive because the command-line parser stores the character exactly as supplied.

/o

Meaning: Controls creation of the remote command communication system.

General form:

```
/o[YES|NO]
```

- YES or an empty value: open remote communications.
- NO (and any other non-YES non-empty value): do not open it.
- Omitted: remote communications disabled.

Examples:

```
C64Emulator.exe /o
C64Emulator.exe /oYES /p60000
```

Consequence: When enabled, remote clients can execute C64, Commodore and standard builder commands. LocalConsole-only commands remain unavailable.

/p

Meaning: Selects the TCP listening port used when /o enables remote communications.

General form:

```
/pPORT
```

- PORT: decimal port number; default 60000 when /p is omitted.

Examples:

```
C64Emulator.exe /o /p60000
C64Emulator.exe /o /p61000
```

Consequence: The communication system attempts to listen on the converted port.

Important: Current-code collision: the inherited Emulator also interprets the same /p value as a comma-separated peripheral ID list. A normal port value can therefore trigger an unintended peripheral connection attempt. Avoid combining startup peripheral attachment with the non-console executable until the option letters are separated in code.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

• No value.

Examples:

```
C64Emulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

• FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
C64Emulator.exe /fprogram.prg
C64Emulator.exe /fgames\demo.prg /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

• FILE: path to a block file understood by the framework.

Examples:

```
C64Emulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

• FILE: assembler source path.

Examples:

```
C64Emulator.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
C64Emulator.exe /l0
C64Emulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: C64 address in the framework number syntax, for example decimal or hexadecimal with $.

Examples:

```
C64Emulator.exe /a$080d
C64Emulator.exe /fprogram.prg /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
C64Emulator.exe /d$080d
C64Emulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
C64Emulator.exe /s
C64Emulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the C64 language/character-ROM variant.

General form:

```
/iLANG
```

- ENG: English (default).
- ESP: Spanish.
- JAP: Japanese.
- SWE: Swedish.
- DKA: Danish.

Examples:

```
C64Emulator.exe /iENG
C64Emulator.exe /iESP
```

Consequence: The selected language is used when constructing the C64 model and its ROM-dependent presentation.

/r

Meaning: Controls sound at startup.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
C64Emulator.exe /r
C64Emulator.exe /rNO
```

Consequence: The audio subsystem begins enabled or disabled. The state can later be changed with SOUNDON/SOUNDOFF.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
C64Emulator.exe /g
C64Emulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Selects NTSC timing instead of PAL.

General form:

```
/n
```

- No value; presence selects NTSC.

Examples:

```
C64Emulator.exe /n
```

Consequence: The emulator creates an NTSC C64. Without /n it creates the PAL model.

/w

Meaning: Selects the SID audio implementation.

General form:

```
/w[OWN|SID]
```

- OWN or an empty value: the framework's ICF/simple SID wrapper (default).
- SID: the reSID-based wrapper.
- Any other text is unsupported and may leave the requested wrapper undefined.

Examples:

```
C64Emulator.exe /wOWN
C64Emulator.exe /wSID
```

Consequence: The selected wrapper is used when the SID chip is created; it changes sound fidelity and processing cost.

/b

Meaning: Draws the C64 screen grid/border using the requested color.

General form:

```
/bCOLOR
```

- COLOR: numeric color value. If /b is present without a valid number, conversion yields 0.

Examples:

```
C64Emulator.exe /b1
C64Emulator.exe /b14 /s
```

Consequence: The screen grid is enabled at startup with the converted color value.

2. Startup Parameters: Version With a Local Console

Executable: C64EmulatorC.exe. Its main program delegates the complete command line to C64::C64Emulator, so all inherited framework options and the C64-specific options are active even though the short main-level help banner only advertises /h. It has no /o listener option; commands are entered in its local console.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
C64Emulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
C64Emulator.exe /fprogram.prg
C64Emulator.exe /fgames\demo.prg /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
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- FILE: path to a block file understood by the framework.

Examples:

```
C64Emulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
C64Emulator.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
C64Emulator.exe /l0
C64Emulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: C64 address in the framework number syntax, for example decimal or hexadecimal with $.

Examples:

```
C64Emulator.exe /a$080d
C64Emulator.exe /fprogram.prg /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
C64Emulator.exe /d$080d
C64Emulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
C64Emulator.exe /s
C64Emulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the C64 language/character-ROM variant.

General form:

```
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```

- ENG: English (default).
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Examples:

```
C64Emulator.exe /iENG
C64Emulator.exe /iESP
```

Consequence: The selected language is used when constructing the C64 model and its ROM-dependent presentation.

/r

Meaning: Controls sound at startup.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
C64Emulator.exe /r
C64Emulator.exe /rNO
```

Consequence: The audio subsystem begins enabled or disabled. The state can later be changed with SOUNDON/SOUNDOFF.

/p

Meaning: Connects framework peripherals during initialization.

General form:

```
/pID1,ID2,...
```

- 10: typewriter input.
- 100: 1530 datasette.
- 101: injected/fast 1530 datasette.
- 203: cartridge.
- 205 or 206: serial printer instances.
- 215, 216, 217 or 218: 1541 disk-drive instances.

Examples:

```
C64EmulatorC.exe /p100
C64EmulatorC.exe /p100,101
```

Consequence: Each recognized ID is built and connected before execution; unsupported IDs are not connected.

Important: Use POSSIBLEPERS in the local console to obtain the IDs supported by the current build.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
C64Emulator.exe /g
```

```
C64Emulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Selects NTSC timing instead of PAL.

General form:

```
/n
```

- No value; presence selects NTSC.

Examples:

```
C64Emulator.exe /n
```

Consequence: The emulator creates an NTSC C64. Without /n it creates the PAL model.

/w

Meaning: Selects the SID audio implementation.

General form:

```
/w[OWN|SID]
```

- OWN or an empty value: the framework's ICF/simple SID wrapper (default).
- SID: the reSID-based wrapper.
- Any other text is unsupported and may leave the requested wrapper undefined.

Examples:

```
C64Emulator.exe /wOWN
C64Emulator.exe /wSID
```

Consequence: The selected wrapper is used when the SID chip is created; it changes sound fidelity and processing cost.

/b

Meaning: Draws the C64 screen grid/border using the requested color.

General form:

```
/bCOLOR
```

- COLOR: numeric color value. If /b is present without a valid number, conversion yields 0.

Examples:

```
C64Emulator.exe /b1
C64Emulator.exe /b14 /s
```

Consequence: The screen grid is enabled at startup with the converted color value.

3. Command Console Reference

The local console checks its eleven emulator-level commands first. All remaining text is delegated through C64::CommandBuilder, then COMMODORE::CommandBuilder, and finally MCHEmul::StandardCommandBuilder. The remote /o channel in C64Emulator.exe uses only that three-builder chain. The availability line in each entry makes this boundary explicit.

3.1 Standard Framework Commands

HELP

Availability: Local console and remote /o channel.

Accepted aliases: CHELP.

Meaning: Displays the C64 console help assembled from the C64, Commodore, local-console and standard help files.

General syntax:

```
HELP
```

Parameters:

- No parameters.

Examples:

```
HELP
```

Result and consequences: Displays the C64 console help assembled from the C64, Commodore, local-console and standard help files.

AUTHOR

Availability: Local console and remote /o channel.

Accepted aliases: CAUTHOR.

Meaning: Returns framework author information.

General syntax:

```
AUTHOR
```

Parameters:

- No parameters.

Examples:

```
AUTHOR
```

Result and consequences: Returns framework author information.

STATUS

Availability: Local console and remote /o channel.

Accepted aliases: CSTATUS.

Meaning: Shows the 6510 status register.

General syntax:

```
STATUS
```

Parameters:

- No parameters.

Examples:

```
STATUS
```

Result and consequences: Shows the 6510 status register.

PC

Availability: Local console and remote /o channel.

Accepted aliases: CPC.

Meaning: Shows the current program counter.

General syntax:

```
PC
```

Parameters:

- No parameters.

Examples:

```
PC
```

Result and consequences: Shows the current program counter.

REGISTERS

Availability: Local console and remote /o channel.

Accepted aliases: CREGISTERS.

Meaning: Shows all CPU registers.

General syntax:

```
REGISTERS
```

Parameters:

- No parameters.

Examples:

```
REGISTERS
```

Result and consequences: Shows all CPU registers.

CHANGEREG

Availability: Local console and remote /o channel.

Accepted aliases: CCHANGEREG.

Meaning: Changes one CPU internal register.

General syntax:

```
CHANGEREG REGNAME VALUE
```

Parameters:

- REGNAME: registered 6510 register name.
- VALUE: new decimal, z-prefixed binary, o-prefixed octal or $-prefixed hexadecimal value.

Examples:

```
CHANGEREG A $41
```

Result and consequences: The register changes only if it exists and accepts the supplied byte width; otherwise an error is returned.

STACK

Availability: Local console and remote /o channel.

Accepted aliases: CSTACK.

Meaning: Shows stack state.

General syntax:

```
STACK [ALL]
```

Parameters:

- ALL: optional; includes stack contents in addition to basic state.

Examples:

```
STACK
STACK ALL
```

Result and consequences: Shows stack state.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATUS.

Meaning: Shows registers, status, program counter and stack.

General syntax:

```
CPUSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSTATUS
```

Result and consequences: Shows registers, status, program counter and stack.

CPUSSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSSTATUS.

Meaning: Shows the compact CPU status without stack detail.

General syntax:

```
CPUSSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSSTATUS
```

Result and consequences: Shows the compact CPU status without stack detail.

CPUSTATE

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATE.

Meaning: Shows the CPU execution state and state-specific data.

General syntax:

```
CPUSTATE
```

Parameters:

- No parameters.

Examples:

```
CPUSTATE
```

Result and consequences: Shows the CPU execution state and state-specific data.

CPUINFO

Availability: Local console and remote /o channel.

Accepted aliases: CCPUINFO.

Meaning: Returns the complete CPU information structure.

General syntax:

```
CPUINFO
```

Parameters:

- No parameters.

Examples:

```
CPUINFO
```

Result and consequences: Returns the complete CPU information structure.

MEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORY.

Meaning: Reads the active CPU-visible memory range.

General syntax:

```
MEMORY INITIAL [FINAL]
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address; with no final address only the initial location is reported.

Examples:

```
MEMORY $0400
MEMORY $0400 $0427
```

Result and consequences: Reads the active CPU-visible memory range.

MEMORYDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYDUMP.

Meaning: Dumps every memory view between two addresses, including inactive RAM/ROM views.

General syntax:

```
MEMORYDUMP INITIAL FINAL
```

Parameters:

- INITIAL: first address.
- FINAL: last address.

Examples:

```
MEMORYDUMP $a000 $a03f
```

Result and consequences: Dumps every memory view between two addresses, including inactive RAM/ROM views.

MEMORYSTR

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYSTR.

Meaning: Shows the configured memory structure.

General syntax:

```
MEMORYSTR [ALL]
```

Parameters:

- ALL: optional; includes inactive regions. Without it only active regions are shown.

Examples:

```
MEMORYSTR
MEMORYSTR ALL
```

Result and consequences: Shows the configured memory structure.

SETMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CSETMEMORY.

Meaning: Writes a value to one address or repeatedly across a range.

General syntax:

```
SETMEMORY INITIAL [FINAL] VALUE
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address.
- VALUE: byte sequence in framework numeric syntax; multi-byte values advance by their own length.

Examples:

```
SETMEMORY $0400 $41
SETMEMORY $0400 $0427 $20
```

Result and consequences: The active memory view is modified immediately; unsafe writes can corrupt state or crash the emulated program.

MEMORYANALYSIS

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYANALYSIS.

Meaning: Analyzes the current memory configuration.

General syntax:

```
MEMORYANALYSIS
```

Parameters:

- No parameters.

Examples:

```
MEMORYANALYSIS
```

Result and consequences: Analyzes the current memory configuration.

STOP

Availability: Local console and remote /o channel.

Accepted aliases: CSTOP.

Meaning: Stops CPU execution.

General syntax:

```
STOP
```

Parameters:

- No parameters.

Examples:

```
STOP
```

Result and consequences: Stops CPU execution.

RUN

Availability: Local console and remote /o channel.

Accepted aliases: R, CRUN.

Meaning: Starts or resumes CPU execution; alias R.

General syntax:

```
RUN [ADDRESS]
```

Parameters:

- ADDRESS: optional new program counter before running.

Examples:

```
RUN
R $080d
```

Result and consequences: The CPU resumes immediately, optionally after changing PC.

SETPC

Availability: Local console and remote /o channel.

Accepted aliases: CSETPC.

Meaning: Sets the program counter without starting execution.

General syntax:

```
SETPC ADDRESS
```

Parameters:

- ADDRESS: new CPU address.

Examples:

```
SETPC $080d
```

Result and consequences: Subsequent execution continues from the new address; an unsafe address can break the program.

NEXT

Availability: Local console and remote /o channel.

Accepted aliases: N, CNEXT.

Meaning: Executes the next instruction; alias N.

General syntax:

```
NEXT
```

Parameters:

- No parameters.

Examples:

```
NEXT
N
```

Result and consequences: Executes the next instruction; alias N.

NEXTSTACK

Availability: Local console and remote /o channel.

Accepted aliases: NS, CNEXTSTACK.

Meaning: Runs until the stack pointer returns to its initial value; alias NS.

General syntax:

```
NEXTSTACK
```

Parameters:

- No parameters.

Examples:

```
NS
```

Result and consequences: Useful for stepping over a routine, although behavior depends on the routine's stack discipline.

SHOWNEXT

Availability: Local console and remote /o channel.

Accepted aliases: SN, CSHOWNEXT.

Meaning: Disassembles upcoming instruction(s); alias SN.

General syntax:

```
SHOWNEXT [NUMBER]
```

Parameters:

- NUMBER: optional count; one instruction by default.

Examples:

```
SHOWNEXT
SN 8
```

Result and consequences: Disassembles upcoming instruction(s); alias SN.

INST

Availability: Local console and remote /o channel.

Accepted aliases: CINST.

Meaning: Shows the last instruction executed.

General syntax:

```
INST
```

Parameters:

- No parameters.

Examples:

```
INST
```

Result and consequences: Shows the last instruction executed.

BREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CBREAKS.

Meaning: Lists all stop-point addresses.

General syntax:

```
BREAKS
```

Parameters:

- No parameters.

Examples:

```
BREAKS
```

Result and consequences: Lists all stop-point addresses.

SETBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CSETBREAK.

Meaning: Adds one or more breakpoints.

General syntax:

```
SETBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more addresses in framework numeric syntax.

Examples:

```
SETBREAK $080d $ffd2
```

Result and consequences: Adds one or more breakpoints.

REMOVEBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAK.

Meaning: Removes selected breakpoints.

General syntax:

```
REMOVEBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more existing breakpoint addresses.

Examples:

```
REMOVEBREAK $ffd2
```

Result and consequences: Removes selected breakpoints.

REMOVEBREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAKS.

Meaning: Removes every breakpoint.

General syntax:

```
REMOVEBREAKS
```

Parameters:

- No parameters.

Examples:

```
REMOVEBREAKS
```

Result and consequences: Removes every breakpoint.

SPEED

Availability: Local console and remote /o channel.

Accepted aliases: CSPEED.

Meaning: Shows measured CPU and screen-I/O execution speed.

General syntax:

```
SPEED
```

Parameters:

- No parameters.

Examples:

```
SPEED
```

Result and consequences: Shows measured CPU and screen-I/O execution speed.

LOADBIN

Availability: Local console and remote /o channel.

Accepted aliases: CLOADBIN.

Meaning: Loads a raw binary file at an explicit address.

General syntax:

```
LOADBIN FILE ADDRESS
```

Parameters:

- FILE: raw binary path.
- ADDRESS: first destination address.

Examples:

```
LOADBIN charset.bin $2000
```

Result and consequences: Bytes are copied into memory without an embedded load-address header.

SAVEBIN

Availability: Local console and remote /o channel.

Accepted aliases: CSAVEBIN.

Meaning: Saves a memory range as raw binary.

General syntax:

```
SAVEBIN FILE SIZE ADDRESS
```

Parameters:

- FILE: output path.
- SIZE: byte count.
- ADDRESS: first source address.

Examples:

```
SAVEBIN screen.bin 1000 $0400
```

Result and consequences: Saves a memory range as raw binary.

DEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CACTIVATEDEEPDEBUG.

Meaning: Enables deep execution debugging.

General syntax:

```
DEBUGON FILE CPU [APPEND] [FILTERS...]
```

Parameters:

- FILE: output log.
- CPU: YES includes CPU execution; NO omits it.
- APPEND: optional YES appends, NO overwrites (default).
- D:START-END or ALLDIR: address filter.
- C:IDs or ALLCHIP: chip filter.
- I:IDs or ALLIO: device filter.
- M:IDs or ALLMEM: memory filter.

Examples:

```
DEBUGON c64.log YES NO D:$0800-$0fff ALLCHIP ALLIO ALLMEM
```

Result and consequences: Tracing begins with the selected sources and can substantially slow emulation. It cannot coexist with interrupt-only debugging.

DEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CDESACTIVATEDEEPDEBUG.

Meaning: Stops deep execution debugging.

General syntax:

```
DEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
DEBUGOFF
```

Result and consequences: Stops deep execution debugging.

IDEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGON.

Meaning: Enables interrupt-only debugging.

General syntax:

```
IDEBUGON FILE [APPEND]
```

Parameters:

- FILE: output log.
- APPEND: optional YES appends; NO overwrites (default).

Examples:

```
IDEBUGON interrupts.log NO
```

Result and consequences: Interrupt tracing begins and cannot coexist with deep debugging.

IDEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGOFF.

Meaning: Stops interrupt-only debugging.

General syntax:

```
IDEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
IDEBUGOFF
```

Result and consequences: Stops interrupt-only debugging.

RESTART

Availability: Local console and remote /o channel.

Accepted aliases: CRESTART.

Meaning: Requests a restart and exits the current computer loop so it can be rebuilt.

General syntax:

```
RESTART STOPPED [LEVEL]
```

Parameters:

- STOPPED: YES restarts stopped; NO restarts running.

• LEVEL: optional 0 CPU-only/mild restart (default), 1 computer restart preserving memory, greater than 1 full restart.

Examples:

```
RESTART YES 0
RESTART NO 2
```

Result and consequences: Requests a restart and exits the current computer loop so it can be rebuilt.

DEVICES

Availability: Local console and remote /o channel.

Accepted aliases: CDEVICES.

Meaning: Shows all or selected I/O devices.

General syntax:

```
DEVICES [ID1 ID2 ...]
```

Parameters:

• IDn: optional device identifiers; with none, all devices are returned.

Examples:

```
DEVICES
DEVICES 1 2
```

Result and consequences: Shows all or selected I/O devices.

PERIPHERALS

Availability: Local console and remote /o channel.

Accepted aliases: CPERIPHERALS.

Meaning: Shows all or selected connected peripherals.

General syntax:

```
PERIPHERALS [ID1 ID2 ...]
```

Parameters:

• IDn: optional peripheral identifiers; with none, all are returned.

Examples:

```
PERIPHERALS
PERIPHERALS 100
```

Result and consequences: Shows all or selected connected peripherals.

PERCOMMAND

Availability: Local console and remote /o channel.

Accepted aliases: CPERCMD.

Meaning: Sends a device-specific command to a connected peripheral.

General syntax:

```
PERCOMMAND PERID CMDID [ATTRS...]
```

Parameters:

- PERID: connected peripheral ID.
- CMDID: command ID implemented by that peripheral.
- ATTRS: optional positional attributes forwarded unchanged.

Examples:

```
PERCOMMAND 100 0
```

Result and consequences: The peripheral executes the request if both IDs and its parameter contract are valid; otherwise the answer reports an error.

ASSIGNJOY

Availability: Local console and remote /o channel.

Accepted aliases: CASSIGNJ.

Meaning: Remaps a joystick identifier.

General syntax:

```
ASSIGNJOY ID NEWID
```

Parameters:

- ID: current joystick ID.
- NEWID: required target ID; using the same ID removes an existing assignment.

Examples:

```
ASSIGNJOY 0 1
```

Result and consequences: Remaps a joystick identifier.

CLOCKFACTOR

Availability: Local console and remote /o channel.

Accepted aliases: CCLOCKFACTOR.

Meaning: Changes the CPU clock multiplier.

General syntax:

```
CLOCKFACTOR VALUE
```

Parameters:

- VALUE: numeric factor; 1 means normal speed.

Examples:

```
CLOCKFACTOR 2
```

Result and consequences: The emulation pacing target changes; values above one request faster execution.

SOUNDON

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDON.

Meaning: Enables audio output.

General syntax:

```
SOUNDON
```

Parameters:

- No parameters.

Examples:

```
SOUNDON
```

Result and consequences: Enables audio output.

SOUNDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDOFF.

Meaning: Disables audio output.

General syntax:

```
SOUNDOFF
```

Parameters:

- No parameters.

Examples:

```
SOUNDOFF
```

Result and consequences: Disables audio output.

INTERRUPTS

Availability: Local console and remote /o channel.

Accepted aliases: CINTERRUPTS.

Meaning: Shows all interrupts known by the computer.

General syntax:

```
INTERRUPTS
```

Parameters:

- No parameters.

Examples:

```
INTERRUPTS
```

Result and consequences: Shows all interrupts known by the computer.

INTSET

Availability: Local console and remote /o channel.

Accepted aliases: CINTSET.

Meaning: Enables or disables one interrupt or every interrupt.

General syntax:

```
INTSET ID|ALL ON|OFF
```

Parameters:

- ID or ALL: target interrupt.
- ON or OFF: requested state.

Examples:

```
INTSET ALL ON
```

Result and consequences: An unknown interrupt ID produces an error.

CHIPS

Availability: Local console and remote /o channel.

Accepted aliases: CCHIPS.

Meaning: Lists the emulated chips and their identifying attributes.

General syntax:

```
CHIPS
```

Parameters:

- No parameters.

Examples:

```
CHIPS
```

Result and consequences: Lists the emulated chips and their identifying attributes.

CRTON

Availability: Local console and remote /o channel.

Accepted aliases: CCRTON.

Meaning: Enables the CRT screen effect.

General syntax:

```
CRTON
```

Parameters:

- No parameters.

Examples:

```
CRTON
```

Result and consequences: Enables the CRT screen effect.

CRTOFF

Availability: Local console and remote /o channel.

Accepted aliases: CCRTOFF.

Meaning: Disables the CRT screen effect.

General syntax:

```
CRTOFF
```

Parameters:

- No parameters.

Examples:

```
CRTOFF
```

Result and consequences: Disables the CRT screen effect.

DATASETTE

Availability: Local console and remote /o channel.

Accepted aliases: CDATASETTE.

Meaning: Shows datasette state when a supported datasette is connected.

General syntax:

```
DATASETTE
```

Parameters:

- No parameters.

Examples:

```
DATASETTE
```

Result and consequences: Shows datasette state when a supported datasette is connected.

GRIDON

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDON.

Meaning: Enables the C64 grid overlay.

General syntax:

```
GRIDON COLOR [ON]
```

Parameters:

- COLOR: numeric grid color.
- ON: optional second token; its presence also enables raster-event and sprite-border markers.

Examples:

```
GRIDON 14
GRIDON 14 ON
```

Result and consequences: The C64-specific implementation shadows the generic standard GRIDON command.

GRIDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDOFF.

Meaning: Disables the C64 grid, raster-event markers and sprite-border markers.

General syntax:

```
GRIDOFF
```

Parameters:

- No parameters.

Examples:

```
GRIDOFF
```

Result and consequences: The C64-specific implementation shadows the generic standard GRIDOFF command.

PICTURE

Availability: Local console and remote /o channel.

Accepted aliases: CPICTURE.

Meaning: Saves a screenshot of the emulator screen.

General syntax:

```
PICTURE FILE
```

Parameters:

- FILE: output PNG, JPG or BMP filename.

Examples:

```
PICTURE c64-screen.png
```

Result and consequences: A supported extension writes the current screen; unsupported formats fail.

SETHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CSETHOOK.

Meaning: Creates an execution hook.

General syntax:

```
SETHOOK ID TYPE [PARAMS...]
```

Parameters:

- ID: unique hook identifier.
- TYPE: hook kind; currently 0 stops on memory read/write.
- For TYPE 0, PARAMS are mandatory INITIAL and optional FINAL addresses.

Examples:

```
SETHOOK 1 0 $d000 $d3ff
```

Result and consequences: Creates an execution hook.

REMOVEHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEHOOK.

Meaning: Removes a hook by identifier.

General syntax:

```
REMOVEHOOK ID
```

Parameters:

- ID: hook identifier.

Examples:

```
REMOVEHOOK 1
```

Result and consequences: If the hook does not exist, no state changes.

HOOKS

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKS.

Meaning: Lists active hooks.

General syntax:

```
HOOKS
```

Parameters:

- No parameters.

Examples:

```
HOOKS
```

Result and consequences: Lists active hooks.

HOOKSHELP

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKSHELP.

Meaning: Shows help for supported hook types.

General syntax:

```
HOOKSHELP
```

Parameters:

- No parameters.

Examples:

```
HOOKSHELP
```

Result and consequences: Shows help for supported hook types.

CMOVEPARAMS

Availability: Local console and remote /o channel.

Accepted aliases: No public alias.

Meaning: Builds the framework's internal MoveParametersToAnswerCommand.

General syntax:

```
CMOVEPARAMS
```

Parameters:

- No parameters are executable under the current command contract.

Examples:

```
CMOVEPARAMS
```

Result and consequences: The command returns an empty answer when invoked directly; it is an internal response helper and is not advertised in the help file.

3.2 Inherited Commodore Commands

VICII

Availability: Local console and remote /o channel.

Accepted aliases: CVICII.

Meaning: Shows VIC-II state, including raster information.

General syntax:

```
VICII
```

Parameters:

- No parameters.

Examples:

```
VICII
```

Result and consequences: Shows VIC-II state, including raster information.

VICIIEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: None in the current internal-name registration.

Meaning: Enables or disables VIC-II event visualization.

General syntax:

```
VICIIEVENTS ON|OFF
```

Parameters:

- ON: enable event display.
- OFF: disable it.

Examples:

```
VICIIEVENTS ON
```

Result and consequences: Enables or disables VIC-II event visualization.

SID

Availability: Local console and remote /o channel.

Accepted aliases: CSID.

Meaning: Shows SID register, voice and wrapper state.

General syntax:

```
SID
```

Parameters:

- No parameters.

Examples:

```
SID
```

Result and consequences: Shows SID register, voice and wrapper state.

SIDW

Availability: Local console and remote /o channel.

Accepted aliases: CSIDW.

Meaning: Changes the active SID wrapper.

General syntax:

```
SIDW LIBID
```

Parameters:

- 0: reSID wrapper.
- 1: ICF/simple wrapper.

Examples:

```
SIDW 0
SIDW 1
```

Result and consequences: The sound implementation changes immediately; other values return an undefined-wrapper error.

VICI

Availability: Local console and remote /o channel.

Accepted aliases: CVICI.

Meaning: Requests VIC-I state.

General syntax:

```
VICI
```

Parameters:

- No parameters.

Examples:

```
VICI
```

Result and consequences: The name is accepted by the inherited Commodore builder, but a C64 has no VIC-I, so the command returns no useful chip data.

TED

Availability: Local console and remote /o channel.

Accepted aliases: CTED.

Meaning: Requests TED state.

General syntax:

```
TED
```

Parameters:

- No parameters.

Examples:

```
TED
```

Result and consequences: Accepted by the inherited builder but not applicable to C64 hardware.

TEDEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: None in the current internal-name registration.

Meaning: Requests TED event visualization.

General syntax:

```
TEDEVENTS ON|OFF
```

Parameters:

- ON or OFF: requested event-display state.

Examples:

```
TEDEVENTS ON
```

Result and consequences: Accepted by the inherited builder but not applicable to C64 hardware.

VIA

Availability: Local console and remote /o channel.

Accepted aliases: CVIA.

Meaning: Requests the generic Commodore VIA state.

General syntax:

```
VIA
```

Parameters:

- No parameters.

Examples:

```
VIA
```

Result and consequences: Accepted by the inherited builder; the C64 uses CIA1/CIA2 instead, so this is not the useful C64 status command.

CIA

Availability: Local console and remote /o channel.

Accepted aliases: CCIA.

Meaning: Requests a generic Commodore CIA state.

General syntax:

```
CIA
```

Parameters:

- No parameters.

Examples:

```
CIA
```

Result and consequences: Accepted by the inherited builder, but C64 exposes its two controllers through CIA1 and CIA2; use those commands for deterministic output.

3.3 C64-Specific Commands

CIA1

Availability: Local console and remote /o channel.

Accepted aliases: CCIA1.

Meaning: Shows CIA #1 state.

General syntax:

```
CIA1
```

Parameters:

- No parameters.

Examples:

```
CIA1
```

Result and consequences: Shows CIA #1 state.

CIA2

Availability: Local console and remote /o channel.

Accepted aliases: CCIA2.

Meaning: Shows CIA #2 state.

General syntax:

```
CIA2
```

Parameters:

- No parameters.

Examples:

```
CIA2
```

Result and consequences: Shows CIA #2 state.

PLA

Availability: Local console and remote /o channel.

Accepted aliases: CPLA.

Meaning: Shows C64 PLA and memory-mapping state.

General syntax:

```
PLA
```

Parameters:

- No parameters.

Examples:

```
PLA
```

Result and consequences: Shows C64 PLA and memory-mapping state.

SCREENDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CSCREENDUMP.

Meaning: Returns current text-screen memory in hexadecimal.

General syntax:

```
SCREENDUMP
```

Parameters:

- No parameters.

Examples:

```
SCREENDUMP
```

Result and consequences: Returns current text-screen memory in hexadecimal.

COLORDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CCOLORDUMP.

Meaning: Returns current color RAM in hexadecimal.

General syntax:

```
COLORDUMP
```

Parameters:

- No parameters.

Examples:

```
COLORDUMP
```

Result and consequences: Returns current color RAM in hexadecimal.

BITMAPDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CBITMAPDUMP.

Meaning: Returns the current VIC-II bitmap data in hexadecimal.

General syntax:

```
BITMAPDUMP
```

Parameters:

- No parameters.

Examples:

```
BITMAPDUMP
```

Result and consequences: Returns the current VIC-II bitmap data in hexadecimal.

SPRITESDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CSPRITESDUMP.

Meaning: Returns raw data for all or selected sprites.

General syntax:

```
SPRITESDUMP [1..8 ...]
```

Parameters:

- Sprite numbers: optional 1 through 8; duplicates are coalesced. With none, all sprites are returned.

Examples:

```
SPRITESDUMP
SPRITESDUMP 1 3 8
```

Result and consequences: Returns raw data for all or selected sprites.

SPRITESDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CSPRITESDRAW.

Meaning: Renders textual drawings of all or selected sprite data.

General syntax:

```
SPRITESDRAW [1..8 ...]
```

Parameters:

- Sprite numbers: optional 1 through 8; duplicates are coalesced.

Examples:

```
SPRITESDRAW 1 2
```

Result and consequences: Monochrome uses space/#; multicolor uses space, OO, XX and ## for the four pixel values.

CHARSDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CCHARSDRAW.

Meaning: Renders textual drawings of all or selected character glyphs.

General syntax:

```
CHARSDRAW [0..255 ...]
```

Parameters:

- Character codes: optional 0 through 255; with none, all are rendered.

Examples:

```
CHARSDRAW 1 2 65
```

Result and consequences: Renders textual drawings of all or selected character glyphs.

PADDLE

Availability: Local console and remote /o channel.

Accepted aliases: CPADDLE.

Meaning: Connects paddles or restores joysticks on C64 control ports.

General syntax:

```
PADDLE ON|OFF [PORT0 PORT1]
```

Parameters:

- ON or OFF: connect or disconnect paddles.
- PORT0/PORT1: optional ports 0 and/or 1; with none, both ports are affected.

Examples:

```
PADDLE ON 0
PADDLE OFF 0 1
```

Result and consequences: Connects paddles or restores joysticks on C64 control ports.

LIGHTPEN

Availability: Local console and remote /o channel.

Accepted aliases: CLIGHTPEN.

Meaning: Enables or disables mouse-driven light-pen emulation.

General syntax:

```
LIGHTPEN ON|OFF
```

Parameters:

- ON: enable.
- OFF: disable.

Examples:

```
LIGHTPEN ON
```

Result and consequences: The mouse moves the light pen and its left button presses it. This implementation can coexist with joysticks, unlike original hardware.

3.4 Local-Console-Only Commands

LOADPRG

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads, assembles and installs an assembler source through the emulator object.

General syntax:

```
LOADPRG FILE
```

Parameters:

- FILE: assembler source path.

Examples:

```
LOADPRG demo.asm
```

Result and consequences: Compilation or loading errors are reported by the local console.

LOADBINARY

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a binary containing its destination address in the file header.

General syntax:

```
LOADBINARY FILE
```

Parameters:

- FILE: header-addressed binary path.

Examples:

```
LOADBINARY demo.prg
```

Result and consequences: Loads a binary containing its destination address in the file header.

LOADBLOCKS

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a block-structured memory file.

General syntax:

```
LOADBLOCKS FILE
```

Parameters:

- FILE: block file path.

Examples:

```
LOADBLOCKS state.blocks
```

Result and consequences: Loads a block-structured memory file.

DECOMPILE

Availability: Local console only.

Accepted aliases: None.

Meaning: Disassembles a memory region.

General syntax:

```
DECOMPILE ADDRESS BYTES
```

Parameters:

- ADDRESS: first address.
- BYTES: approximate byte count; output rounds to complete instructions.

Examples:

```
DECOMPILE $080d 64
```

Result and consequences: Disassembles a memory region.

POSSIBLEPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Lists peripheral IDs and required attributes supported by the active I/O peripheral builder.

General syntax:

```
POSSIBLEPERS
```

Parameters:

- No parameters.

Examples:

```
POSSIBLEPERS
```

Result and consequences: Lists peripheral IDs and required attributes supported by the active I/O peripheral builder.

CONNECTPER

Availability: Local console only.

Accepted aliases: None.

Meaning: Builds and connects a peripheral to the first compatible device.

General syntax:

```
CONNECTPER ID [ATTRS...]
```

Parameters:

- ID: peripheral-emulation ID from POSSIBLEPERS.
- ATTRS: optional device-specific positional attributes, such as d:8 for a 1541 or d:4 f:printer.ps p:MPS801-PS for a printer.

Examples:

```
CONNECTPER 100
CONNECTPER 215 d:8
```

Result and consequences: Connection succeeds only when the ID and attributes are valid and a compatible device is available.

DISCONNECTPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Disconnects one or more peripherals.

General syntax:

```
DISCONNECTPERS ID1 [ID2 ...]
```

Parameters:

- IDn: connected peripheral IDs.

Examples:

```
DISCONNECTPERS 100 101
```

Result and consequences: Disconnects one or more peripherals.

LOADPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads media/data into a connected peripheral.

General syntax:

```
LOADPERDATA ID FILE [DEBUGFILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: supported media/data file.
- DEBUGFILE: optional file used to start format-related debugging.

Examples:

```
LOADPERDATA 215 disk.d64
```

Result and consequences: The operation fails if the peripheral does not support the supplied format.

EMPTYPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Creates empty media/data in a peripheral.

General syntax:

```
EMPTYPERDATA ID FILE
```

Parameters:

- ID: connected peripheral ID.
- FILE: path whose extension selects a supported creatable format.

Examples:

```
EMPTYPERDATA 215 blank.d64
```

Result and consequences: Only peripherals and formats that implement empty-media creation can succeed.

SAVEPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Saves a peripheral's current media/data.

General syntax:

```
SAVEPERDATA ID [FILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: optional output path; may be omitted when LOADPERDATA or EMPTYPERDATA established the current filename.

Examples:

```
SAVEPERDATA 215
SAVEPERDATA 215 copy.d64
```

Result and consequences: Saves a peripheral's current media/data.

CLEARPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Clears all data currently loaded into peripherals.

General syntax:

```
CLEARPERDATA
```

Parameters:

- No parameters.

Examples:

```
CLEARPERDATA
```

Result and consequences: Commonly used to force a subsequent program/media reload.

Appendices

Appendix A. Command Availability Summary

The table distinguishes the interactive console from the command channel of the non-console executable. Remote availability assumes C64Emulator.exe was started with /o and that the client is connected.

Command family	C64EmulatorC local console	C64Emulator remote /o
Standard builder	Yes	Yes
Inherited Commodore builder	Yes	Yes
C64 builder	Yes	Yes
LocalConsole emulator/media operations	Yes	No

Appendix B. Source-of-Truth Maintenance Rule

This file is a maintained repository artifact. Every change that adds, removes, renames or changes a C64 startup option, a command-builder registration, a LocalConsole-only command, command syntax, parameter meaning, command consequence or user-visible formatted answer must include a review and, when applicable, an update of docs/C64Emulator_UserGuide.docx. The corresponding canonical .fmt source must be updated whenever the command's InfoStructure contract changes.

Audit the complete chain, not only the edited class: C64Emulator entry point and C64::C64Emulator for startup options; LocalConsole for local-only commands; and C64::CommandBuilder -> COMMODORE::CommandBuilder -> MCHEmul::StandardCommandBuilder for builder commands.

Illustrations

NO ILLUSTRATIONS ARE INCLUDED IN THIS EDITION.

Equations

NO EQUATIONS ARE INCLUDED IN THIS EDITION.